CPE412 Pattern Recognition

Week 10

Linear Regression Analysis – Least Square

Dr. Nehad Ramaha,
Computer Engineering Department
Karabük Universities

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Regression –Problem Formulation

Week 8

- For given example the Linear Regression is modeled as:

$$BloodPressure(y) = w_0 + w_1 AgeinYear(X)$$

OR

$$y = w_0 + w_1 X$$ – Equation of line

with w_0 *is intercept on Y_axis and* w_1 *is slope of line*

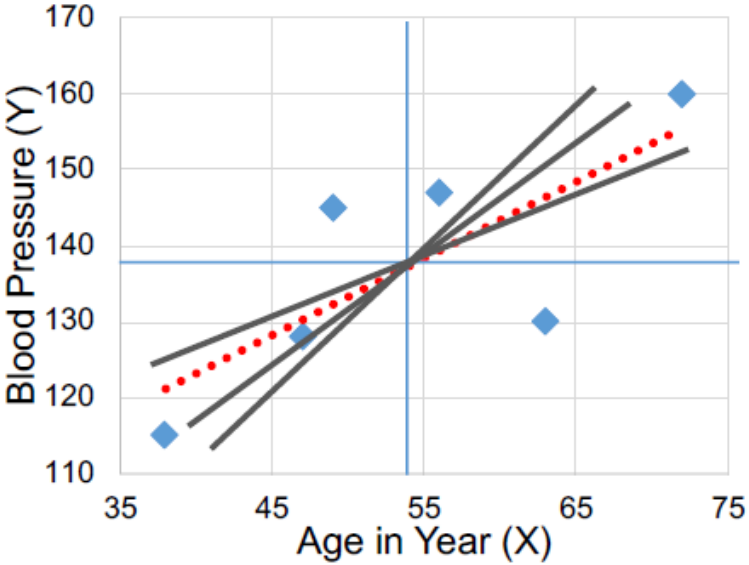
Blood Pressure - **Dependent Variable**

Age in Year - **Independent Variable**

Linear Regression- Best Fit Line

Week 8

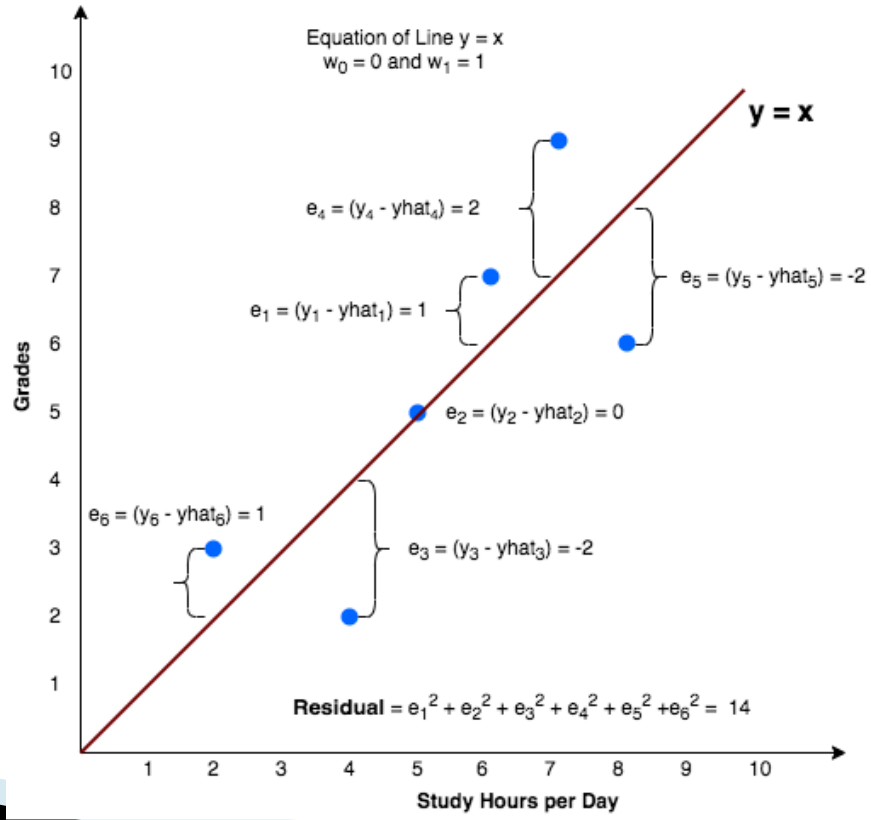
- Regression uses line to show the trend of distribution.
- There can be many lines that try to fit the data points in scatter diagram
- The aim is to find Best fit Line



What is Best Fit Line

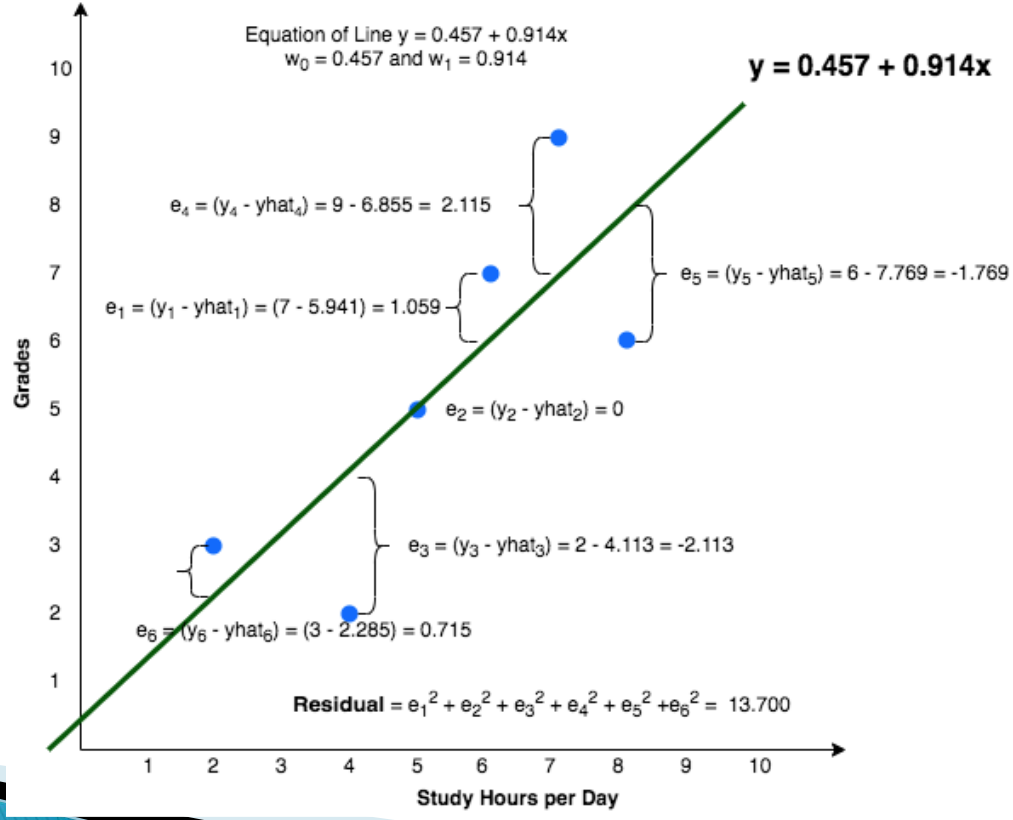
Week 8

- Best fit line tries to explain the variance in given data. (minimize the total residual/error)



What is Best Fit Line

Week 8



Linear Regression- Methods to Get Best

- Least Square
- Gradient Descent

Linear Regression- Least Square

- The line formula $Y = a + b * X$.
- Least squares: is a method to apply linear regression.
- It helps us predict results based on an existing set of data as well as clear anomalies in our data.
- Anomalies are values that are too good, or bad, to be true or that represent rare cases.

Least Square- Example

- Let's assume that our objective is to figure out how many topics are covered by a student per hour of learning.
- Each pair (X, Y) will represent a student. Since we all have different rates of learning, the number of topics solved can be higher or lower for the same time invested.

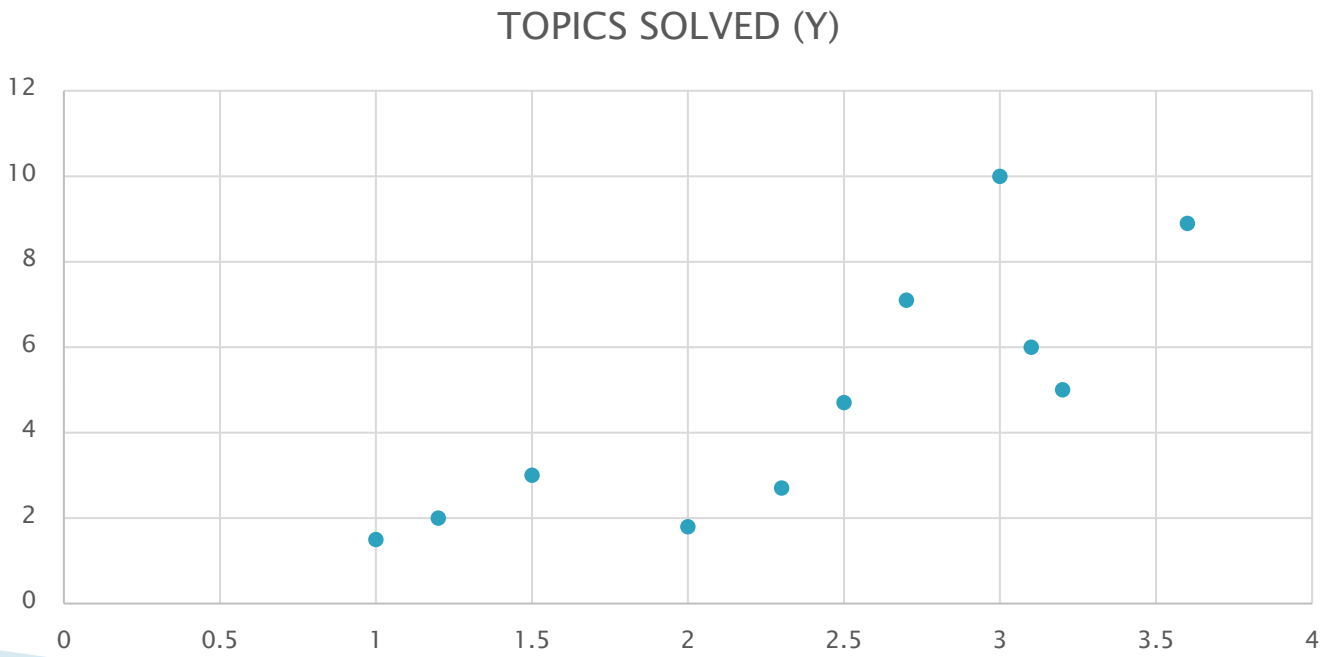
Least Square- Example

- You can read it like this: "Someone spent 1 hour and solved 2 topics" or "One student after 3 hours solved 10 topics".

HOURS (X)	TOPICS SOLVED (Y)
1	1.5
1.2	2
1.5	3
2	1.8
2.3	2.7
2.5	4.7
2.7	7.1
3	10
3.1	6
3.2	5
3.6	8.9

Least Square– Example

In a graph these points look like this: Each point is a student (X, Y) and how long it took that specific student to complete a certain number of topics



Least Square– Example

- The formula: $Y = a + bX$
- we need to figure out the a and b values.
- To give some context as to what they mean:
  - a is the intercept, in other words the value that we expect, on average, from a student that practices for one hour. One hour is the least amount of time we're going to accept into our example data set.
  - b is the slope or coefficient, in other words the number of topics solved in a specific hour (X). As we increase in hours (X) spent studying, b increases more and more.

Least Square– Example

- Calculating "b“:

$$b = \frac{\sum(x - \bar{x}) * (y - \bar{y})}{\sum(x - \bar{x})^2}$$

- Looks scarier than it is
- X and Y are our positions from our earlier table. When they have a – (macron) above them, it means we should use the average which we obtain by summing them all up and dividing by the total amount

Least Square- Example

- $\bar{x}$ -> 1+1.2+1.5+2+2.3+2.5+2.7+3+3.1+3.2+3.6/11 = 2.37
- $\bar{y}$ -> 1.5+2+3+1.8+2.7+4.7+7,1+10+6+5+8.9 / 11 = 4.79

Least Square- Example

- Now that we have the average we can expand our table to include the new results:

HOURS (X)	TOPICS SOLVED (Y)	$(X - \bar{X})$	$(Y - \bar{Y})$	$(X - \bar{X})*(Y - \bar{Y})$	$(X - \bar{X})^2$
1	1.5	-1.37	-3.29	4.51	1.88
1.2	2	-1.17	-2.79	3.26	1.37
1.5	3	-0.87	-1.79	1.56	0.76
2	1.8	-0.37	-2.99	1.11	0.14
2.3	2.7	-0.07	-2.09	0.15	0.00
2.5	4.7	0.13	-0.09	-0.01	0.02
2.7	7.1	0.33	2.31	0.76	0.11
3	10	0.63	5.21	3.28	0.40
3.1	6	0.73	1.21	0.88	0.53
3.2	5	0.83	0.21	0.17	0.69
3.6	8.9	1.23	4.11	5.06	1.51

Least Square- Example

- The weird symbol sigma (Σ) tells us to sum everything up:
- $\Sigma(x - \bar{x})*(y - \bar{y})$ -> 4.51+3.26+1.56+1.11+0.15+-0.01+0.76+3.28+0.88+0.17+5.06 = 20.73
- $\Sigma(x - \bar{x})^2$ -> 1.88+1.37+0.76+0.14+0.00+0.02+0.11+0.40+0.53+0.69+1.51 = 7.41
- And finally we do 20.73 / 7.41 and we get b = 2.8

Least Square- Example

- **Calculating "a"**
- All that is left is a, for which the formula is $\bar{y} = a + b\,\bar{x}$.
- We've already obtained all those other values, so we can substitute them and we get:
- $4.79 = a + 2.8*2.37$
- $4.79 = a + 6.64$
- $a = -6.64+4.79$
- $a = -1.85$

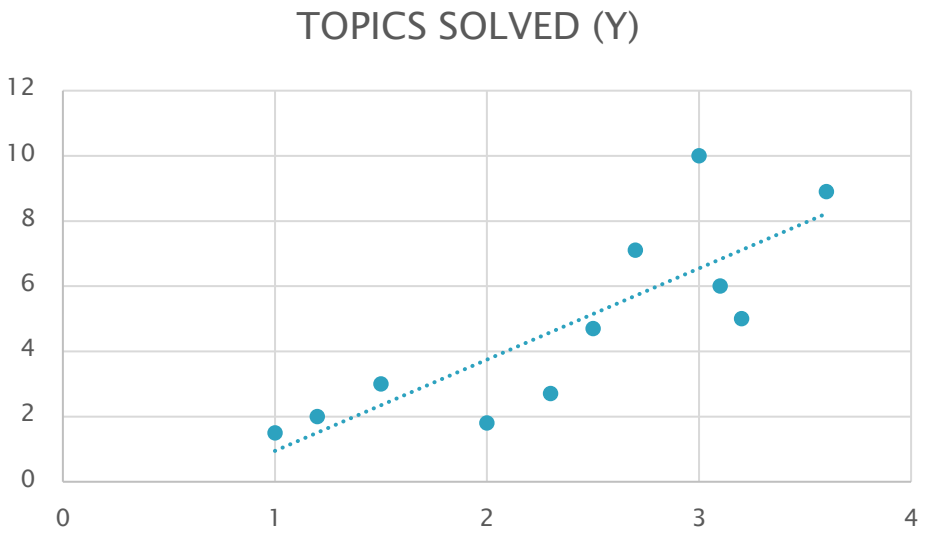
Least Square- Example- The result

- Our final formula becomes:
  $Y = -1.85 + 2.8*X$
- Now we replace the X in our formula with each value that we have:

HOURS (X)	-1.85 + 2.8 * X
1	0.95
1.2	1.51
1.5	2.35
2	3.75
2.3	4.59
2.5	5.15
2.7	5.71
3	6.55
3.1	6.83
3.2	7.11
3.6	8.23

Least Square- Example- The result

- We now have a line that represents how many topics we expect to be solved for each hour of study



Least Square- Example- The result

- If we want to predict how many topics we expect a student to solve with 8 hours of study, we replace it in our formula:

- $Y = -1.85 + 2.8*8$
- $Y = 20.55$

Linear Regression- Example2

- The price and page numbers of the books are given in the table below. Find the regression line (Estimation Equation) by which we can estimate the price of the book using the number of pages.

NUMBER OF PAGES (X)	SALES PRICE (Y)
100	8
150	12
200	14
250	15
300	16

Linear Regression- Example2

Linear Regression- Example2

- Calculating "b“:

$$b = \frac{\sum(x - \bar{x}) * (y - \bar{y})}{\sum(x - \bar{x})^2}$$

- $\bar{x}$ -> 100+150+200+250+300/5 = 200
- $\bar{y}$ -> 8+12+14+15+16/5 = 13

Linear Regression- Example2

- Now that we have the average we can expand our table to include the new results:

NUMBER OF PAGES (X)	SALES PRICE (Y)	$(X - \bar{X})$	$(Y - \bar{Y})$	$(X - \bar{X})*(Y - \bar{Y})$	$(X - \bar{X})^2$
100	8	-100	-5	500	10000
150	12	-50	-1	50	2500
200	14	0	1	0	0
250	15	50	2	100	2500
300	16	100	3	300	10000

Linear Regression- Example2

- The weird symbol sigma (Σ) tells us to sum everything up:
- $\Sigma(x - \overline{x})*(y - \overline{y})$ -> 500 + 50 + 0 +100+300 =950
- $\Sigma(x - \overline{x})^2$ -> 10000+2500+0+2500+10000 =25000
- And finally we do 950 / 25000 and we get b = 0.038

Linear Regression- Example2

- **Calculating "a"**
- All that is left is a, for which the formula is $\bar{y} = a + b\,\bar{x}$.
- We've already obtained all those other values, so we can substitute them and we get:
- $13 = a + 0.038*200$
- $13 = a + 7.6$
- $a = 13-7.6$
- $a = 5.4$

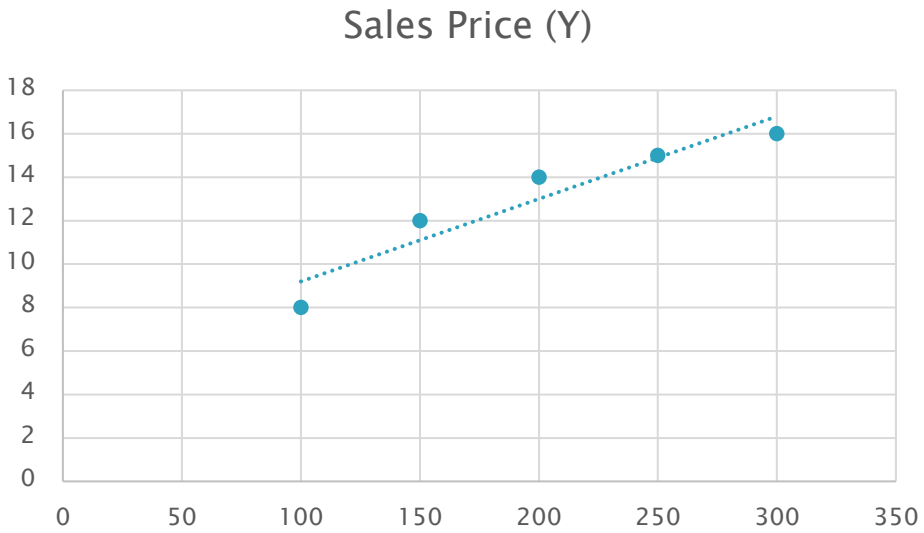
Linear Regression- Example2

- Our final formula becomes:
  Y = 5. 4+ 0.038*X
- Now we replace the X in our formula with each value that we have:

NUMBER OF PAGES (X)	SALES PRICE (Y)
100	9.2
150	11.1
200	13
250	14.9
300	16.8

Linear Regression- Example2

- We now have a line that represents the expected sales price



Linear Regression- Example2

- If we want to predict how much the sale price we expect for 1000 pages book, we replace it in our formula:(y=5.4+0.038*X)

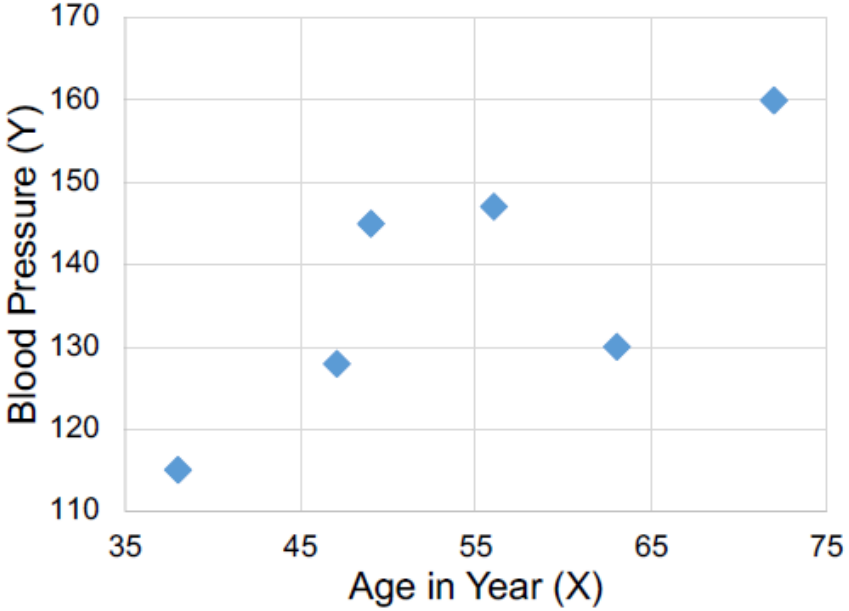
- Y = 5.4+ 0.038*1000
- Y = 43.4 TL

Linear Regression- Example3

- Do it yourself- find the expected Blood pressure for 80 years age?

Let you have given with a data:

Age in Years (X)	Blood Pressure (Y)
56	147
49	145
72	160
38	115
63	130
47	128



Thanks ☺